

Experimental Methods in Computer Science

Departamento de Engenharia Informática, FCTUC, 2025/2026

Experimental Methods in Computer Science (Metodologias Experimentais em Informática)

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Why do we need experiments?



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Why do we need experiments?

Researchers:

- Collect evidence facts about the world (or system)
- Validate hypothesis
- Support the definition, validation, parameterization of models
- Validate models
- Confirm theories
- Etc, etc...

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Why do we need experiments?

Engineers (including informatics engineers):

- Tune up systems
- Compare and select among different project choices
- **Verify** that requirements or specifications are met
- **Validate** mechanisms and/or solutions
- Measure/evaluate features, e.g., to access efficiency of mechanisms.
- Assess the effectiveness of processes, e.g., software development processes
- Etc, etc...

“Experimentation as the feedback step in the engineering loop”

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Key properties

- **Relevance**
Are the goals of the experiment and the expected results important for the progress (do they have impact in science, technology, market, etc.)?
- **Representativeness**
Is the experiment realistic and representative of real-world scenarios?
- **Repeatability**
Is it possible to repeat the experiment and achieve same or statistically similar results?
- **Reproducibility**
Is there enough information to allow others to reproduce the experiment?
- **Results analysis and generalization**
Is the analysis of results sound? Is the generalization of conclusions credible?
- **Cost**
Is the cost of the experiments compatible with the expected benefits?

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Many types of experiments...

- Controlled experiments
- Surveys
- Field studies
- Rational reconstructions
- Case studies
- Artifact/archive analysis
- Pilot studies
- Ethnographies
- Benchmarks
- Quasi-experiments
- Simulations

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Design of experiments (a first look)

Laboratory experiments, controlled experiments

1. Problem statement (or research question)
 2. Identify variables
 3. Generate hypothesis
 4. Define the experimental setup/scenario
 5. Develop tools and procedures for the experiment
 6. Run experiments and collect the data/measurements
 7. Perform data analysis and test hypothesis
 8. Draw conclusions (often go back to the beginning and reformulate the problem statement or test a different hypothesis)
- Design of the experiment
- Measurements
- Analysis (statistics, in practice)
- Conclusions

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- A good (i.e., relevant) problem statement should be focused enough to allow the clear identification of the variables of the problem but, at the same time, should be sufficiently open to allow different hypothesis to answer the problem/question.
- Possible generic formulation:
- How does X affect Y under conditions Z?**

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To formulate good problem statements:

- Must know the subject area: process, system, technique, product, product market, etc.
- Must be precise and clear
- Must be sure that the problem/question is relevant

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• **Dependent variable** (response variable)

Measured output (e.g., response time, throughput, no. bugs, downtime, latency, error detection coverage, etc., etc.)

• **Independent variables** (factors)

Input variables that can be changed in the experiment (e.g., memory size, clock rate, file size, channel bandwidth, etc., etc.)

• **Levels**

Values taken by the variables. Can be (nearly) continuous (e.g., ~time, size in bytes) or discrete (type of system, type of algorithm, etc.)

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- **Change one factor at the time**

Simple scenario. The analysis is simple, as it is easy to understand the effect of a given factor on the independent variable.

- **Full factorial**

Change two or more factors simultaneously. Much more complex (to be seen in detail later on in the course). Has two important advantages:

- More efficient experiments (save time and effort)
- Allow the study of possible interactions among factors

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- Hypothesis describe provisional relationships between factors (independent variables) and the response variable (dependent). **It is an interim answer to the problem statement.**

- Can be directional or non-directional

- A hypothesis is always pair:

- **Null hypothesis (H_0)**

States that there is no effect, no difference, or no relationship between the variables being studied.

- **Alternate hypothesis (H_1)**

States that there is an effect, a difference, or a relationship.

- Example

- **H_0** - There is no difference between the execution time of algorithm A and B

- **H_1** - The execution time of algorithm A is shorter than B

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- Experiment complexity
 - Experiment cost
 - Availability of tools and frameworks that may help
 - Degree of automation

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- Continuous and/or discrete measurements
 - Accuracy, precision, and resolution
 - Basic measurements in computers...
 - Count
 - Duration
 - Size
 - Any value derived from the combination of basic measurements

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Example

Consider the following **question**: Is the number of software bugs found in the tests of program units developed by programmers dependent on the average number of sleeping hours of the programmers?

Assume that you have the detailed specifications of a set of program units to be developed and consider that the program units include units of high, medium and low complexity. Additionally, you have comprehensive unit test suits to test each program unit.

In these circumstances, describe how you would organize an experiment to answer the proposed question (problem statement). Your answer should be as complete as possible, focusing on the experiment design steps (obviously, it does not make sense to speculate about the experiment results and conclusions), and indicate at least the following:

- a) Dependent and independent variables.
- b) Levels you would consider for the independent variables.
- c) Hypothesis under evaluation.